

Parametric Equations and Their Graphs

Parametric Equations: Equations used to define the x and y values of a graph. The independent variable used to define the x and y values is usually " t " and is called the parameter. If " f " is the function used to define x , and " g " is the function used to define y , then $(x, y) = (f(t), g(t))$.

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

① $x = t + 1, y = t^2$

Point #1 $t = 0$
 $x = 0 + 1, y = 0^2$
 $x = 1, y = 0$
 $(1, 0)$

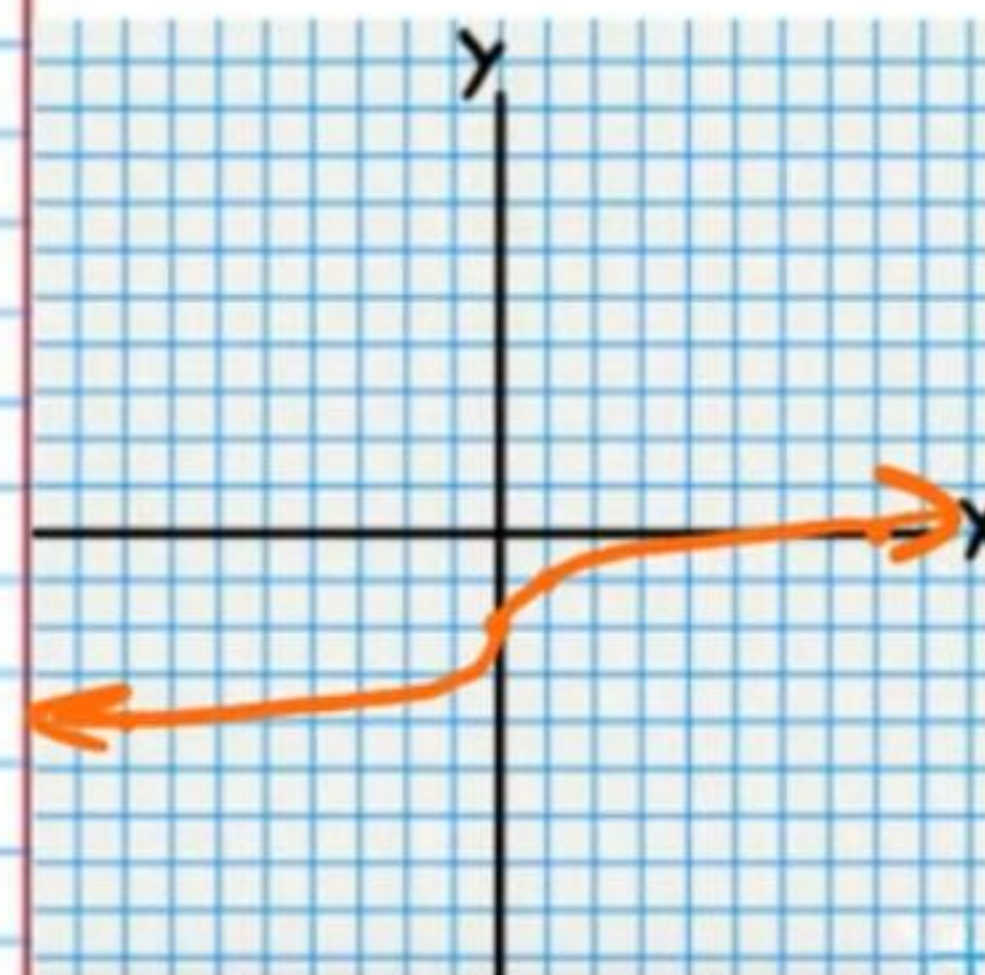
Point #2 $t = 1$
 $x = 1 + 1, y = 1^2$
 $x = 2, y = 1$
 $(2, 1)$

Point #3 $t = -1$
 $x = -1 + 1, y = (-1)^2$
 $x = 0, y = 1$
 $(0, 1)$

Point #4 $t = -2$
 $x = -2 + 1, y = (-2)^2$
 $x = -1, y = 4$
 $(-1, 4)$

Eliminate the parameter
 $x = t + 1, y = t^2$
 $t = x - 1$
 $y = (x - 1)^2$

② $x = t^3, y = t - 2$



Point #1

$t = 0$
 $x = 0^3, y = 0 - 2$
 $x = 0, y = -2$
 $(0, -2)$

Point #2

$t = 1$
 $x = 1^3, y = 1 - 2$
 $x = 1, y = -1$
 $(1, -1)$

Point #3

$t = -2$
 $x = (-2)^3, y = -2 - 2$
 $x = -8, y = -4$
 $(-8, -4)$

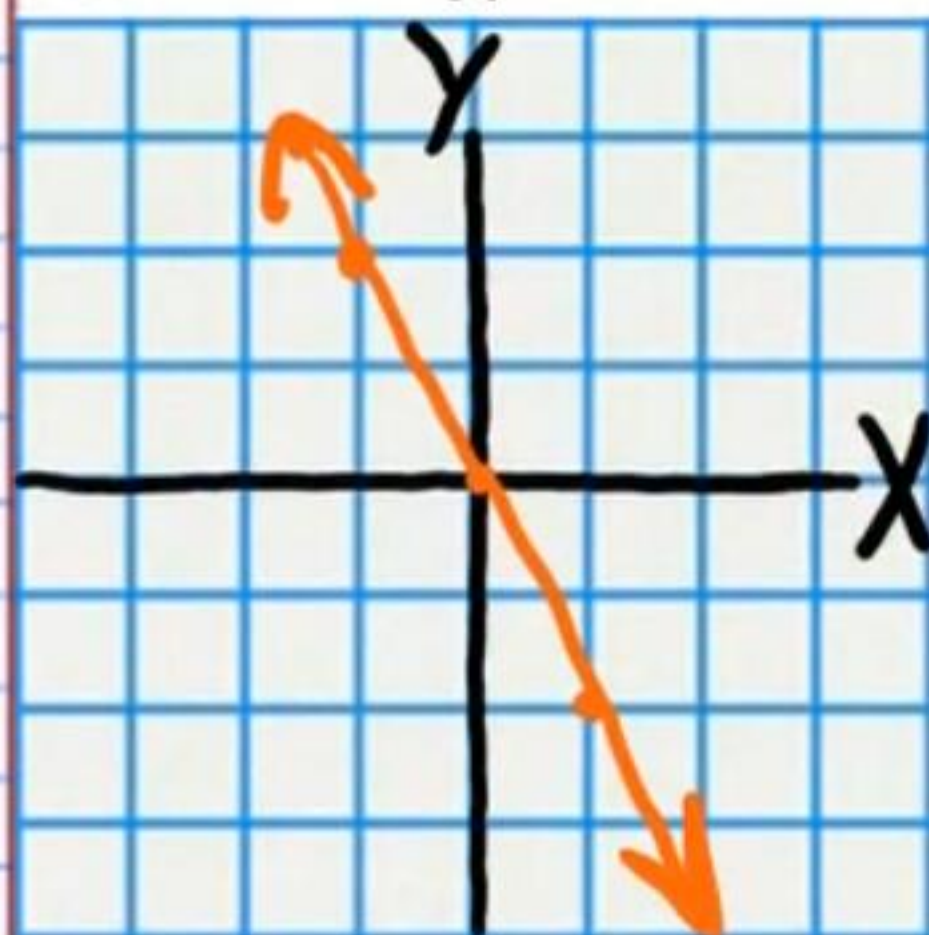
Point #4

$t = 2$
 $x = 2^3, y = 2 - 2$
 $x = 8, y = 0$
 $(8, 0)$

Eliminate the parameter

$x = t^3, y = t - 2$
 $t = y + 2$
 $x = (y + 2)^3$

③ $x = -\frac{t}{2}, y = t$



Point #1

$$\begin{aligned} t &= 0 \\ x &= -\frac{0}{2}, y = 0 \\ x &= 0, y = 0 \\ \boxed{(0, 0)} \end{aligned}$$

Point #2

$$\begin{aligned} t &= -2 \\ x &= -(-\frac{2}{2}), y = -2 \\ x &= 1, y = -2 \\ \boxed{(1, -2)} \end{aligned}$$

Point #4

$$\begin{aligned} t &= 3 \\ x &= -\frac{3}{2}, y = 3 \\ \boxed{(-\frac{3}{2}, 3)} \end{aligned}$$

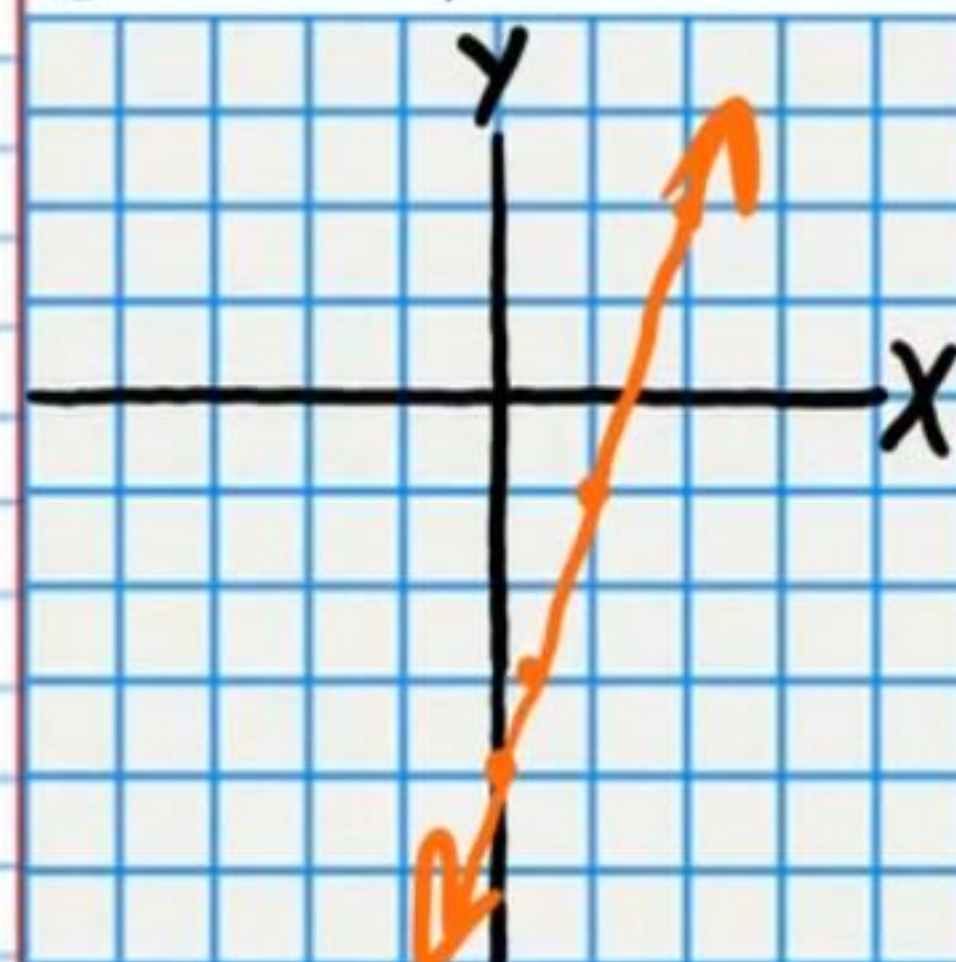
Point #3

$$\begin{aligned} t &= 2 \\ x &= -\frac{2}{2}, y = 2 \\ x &= -1, y = 2 \\ \boxed{(-1, 2)} \end{aligned}$$

Eliminate the parameter

$$\begin{aligned} x &= -\frac{t}{2}, y = t \\ \downarrow \\ \boxed{x = -\frac{y}{2}} \end{aligned}$$

④ $x = t, y = 3t - 4$



Point #1

$$\begin{aligned} t &= 0 \\ x &= 0, y = 3(0) - 4 \\ x &= 0, y = -4 \\ \boxed{(0, -4)} \end{aligned}$$

Point #2

$$\begin{aligned} t &= 1 \\ x &= 1, y = 3(1) - 4 \\ x &= 1, y = -1 \\ \boxed{(1, -1)} \end{aligned}$$

Point #4

$$\begin{aligned} t &= \frac{1}{3} \\ x &= \frac{1}{3}, y = 3(\frac{1}{3}) - 4 \\ x &= \frac{1}{3}, y = 1 - 4 \\ \boxed{(\frac{1}{3}, -3)} \end{aligned}$$

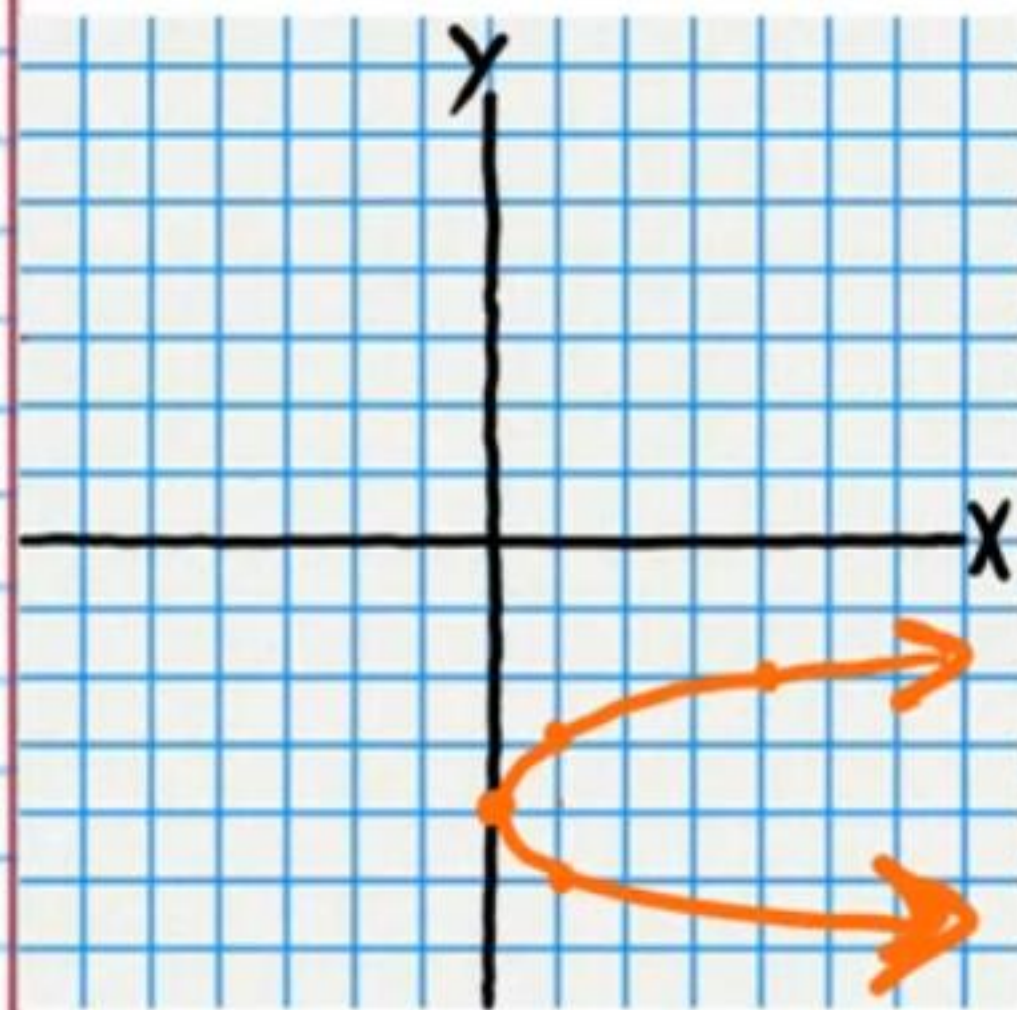
Point #3

$$\begin{aligned} t &= 2 \\ x &= 2, y = 3(2) - 4 \\ x &= 2, y = 2 \\ \boxed{(2, 2)} \end{aligned}$$

Eliminate the parameter

$$\begin{aligned} x &= t, y = 3t - 4 \\ \downarrow \\ \boxed{y = 3x - 4} \end{aligned}$$

$$\textcircled{5} \quad x = t^2, \quad y = t - 4$$



Point #1

$$\begin{aligned} t &= 0 \\ x &= 0^2, \quad y = 0 - 4 \\ x &= 0, \quad y = -4 \\ \boxed{(0, -4)} \end{aligned}$$

Point #2

$$\begin{aligned} t &= 1 \\ x &= 1^2, \quad y = 1 - 4 \\ x &= 1, \quad y = -3 \\ \boxed{(1, -3)} \end{aligned}$$

Point #3

$$\begin{aligned} t &= -1 \\ x &= (-1)^2, \quad y = -1 - 4 \\ x &= 1, \quad y = -5 \\ \boxed{(1, -5)} \end{aligned}$$

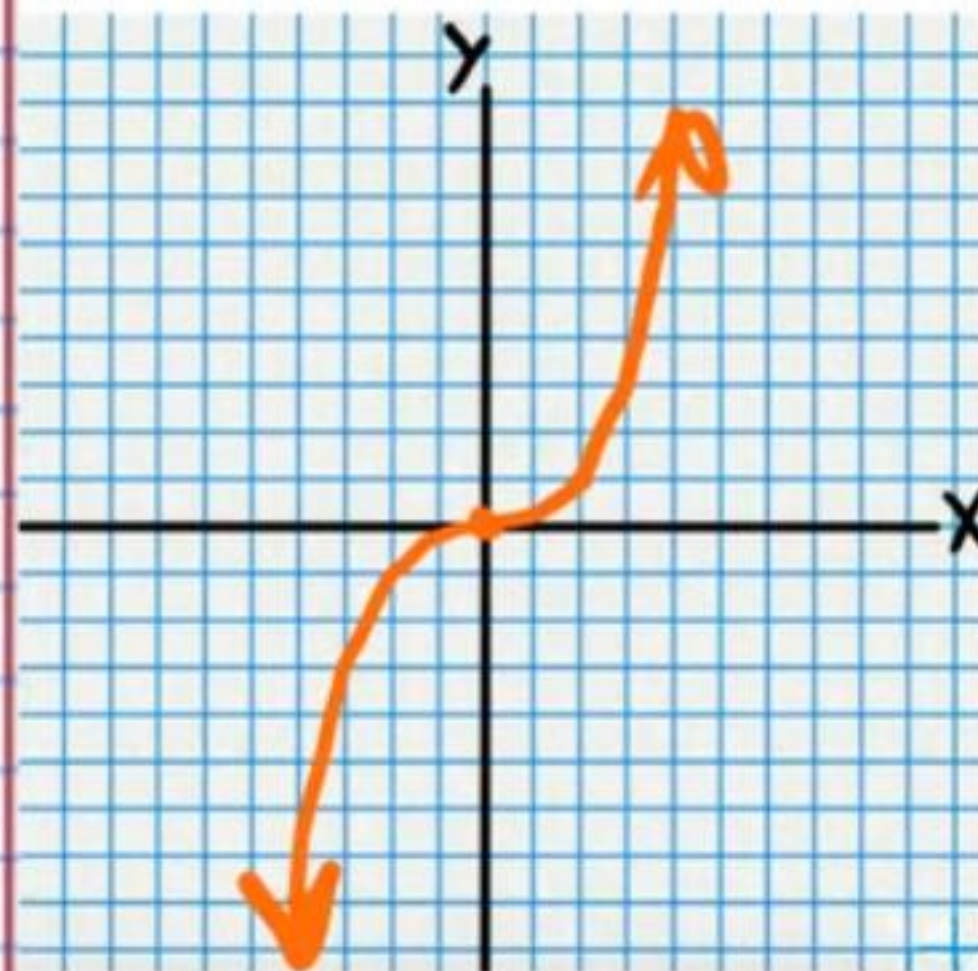
Point #4

$$\begin{aligned} t &= 2 \\ x &= 2^2, \quad y = 2 - 4 \\ x &= 4, \quad y = -2 \\ \boxed{(4, -2)} \end{aligned}$$

Eliminate the parameter

$$\begin{aligned} x &= t^2, \quad y = t - 4 \\ &\downarrow \\ \boxed{x = (y + 4)^2} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} t = y + 4 \end{aligned}$$

$$\textcircled{6} \quad x = 2t, \quad y = t^3$$



Point #1

$$\begin{aligned} t &= 0 \\ x &= 2(0), \quad y = 0^3 \\ x &= 0, \quad y = 0 \\ \boxed{(0, 0)} \end{aligned}$$

Point #2

$$\begin{aligned} t &= 2 \\ x &= 2(2), \quad y = 2^3 \\ x &= 4, \quad y = 8 \\ \boxed{(4, 8)} \end{aligned}$$

Point #4

$$\begin{aligned} t &= 1 \\ x &= 2(1), \quad y = 1^3 \\ x &= 2, \quad y = 1 \\ \boxed{(2, 1)} \end{aligned}$$

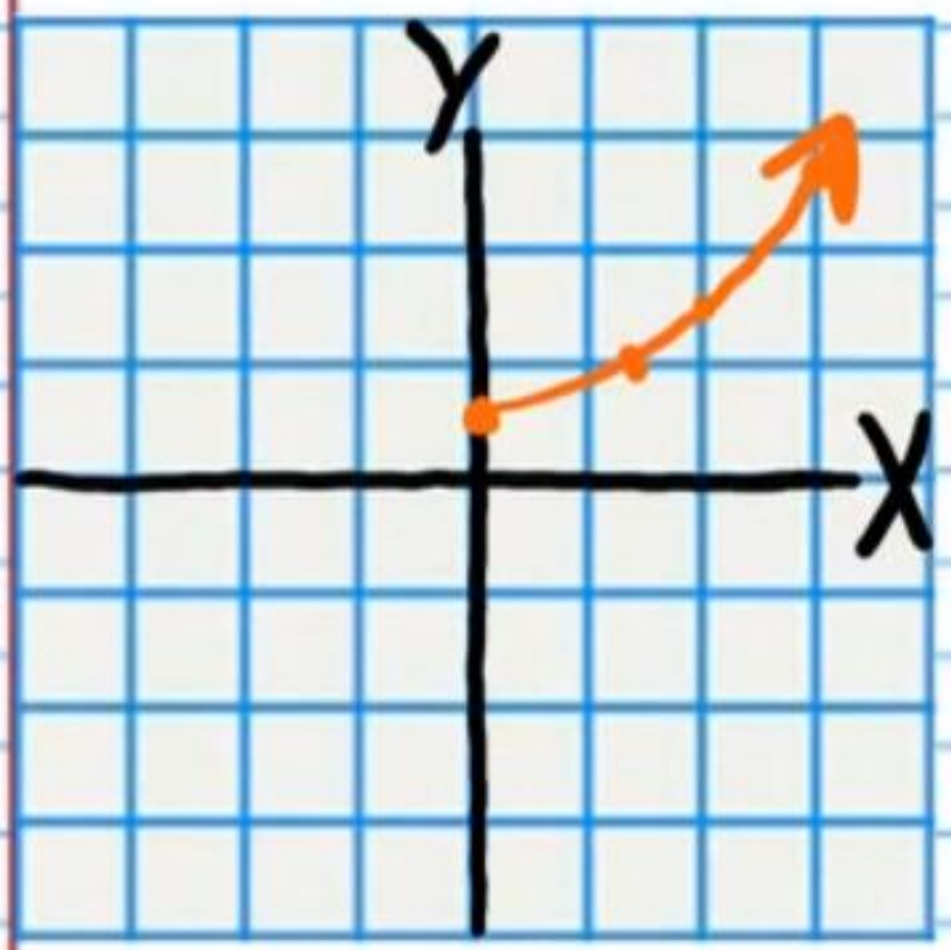
Point #3

$$\begin{aligned} t &= -2 \\ x &= 2(-2), \quad y = (-2)^3 \\ x &= -4, \quad y = -8 \\ \boxed{(-4, -8)} \end{aligned}$$

Eliminate the parameter

$$\begin{aligned} x &= 2t, \quad y = t^3 \\ &\downarrow \quad \quad \downarrow \\ t &= \frac{x}{2} \quad \left\{ \begin{array}{l} y = \left(\frac{x}{2}\right)^3 \\ \boxed{y = \frac{x^3}{8}} \end{array} \right. \end{aligned}$$

⑦ $x = \sqrt{t-2}, y = \frac{t}{4}$



Point #1

$$\begin{aligned} t &= 2 \\ x &= \sqrt{2-2}, y = \frac{2}{4} \\ x &= \sqrt{0}, y = \frac{1}{2} \\ x &= 0, y = \frac{1}{2} \end{aligned}$$

$(0, \frac{1}{2})$

Point #2

$$\begin{aligned} t &= 4 \\ x &= \sqrt{4-2}, y = \frac{4}{4} \end{aligned}$$

$x = \sqrt{2}, y = 1$

$(\sqrt{2}, 1)$

Point #4

$$\begin{aligned} t &= 11 \\ x &= \sqrt{11-2}, y = \frac{11}{4} \end{aligned}$$

$(3, 2\frac{3}{4})$

Point #1

$$\begin{aligned} t &= 2 \\ x &= \sqrt{2-2}, y = \frac{2}{4} \\ x &= \sqrt{0}, y = \frac{1}{2} \\ x &= 0, y = \frac{1}{2} \end{aligned}$$

$(0, \frac{1}{2})$

Point #3

$$\begin{aligned} t &= 6 \\ x &= \sqrt{6-2}, y = \frac{6}{4} \end{aligned}$$

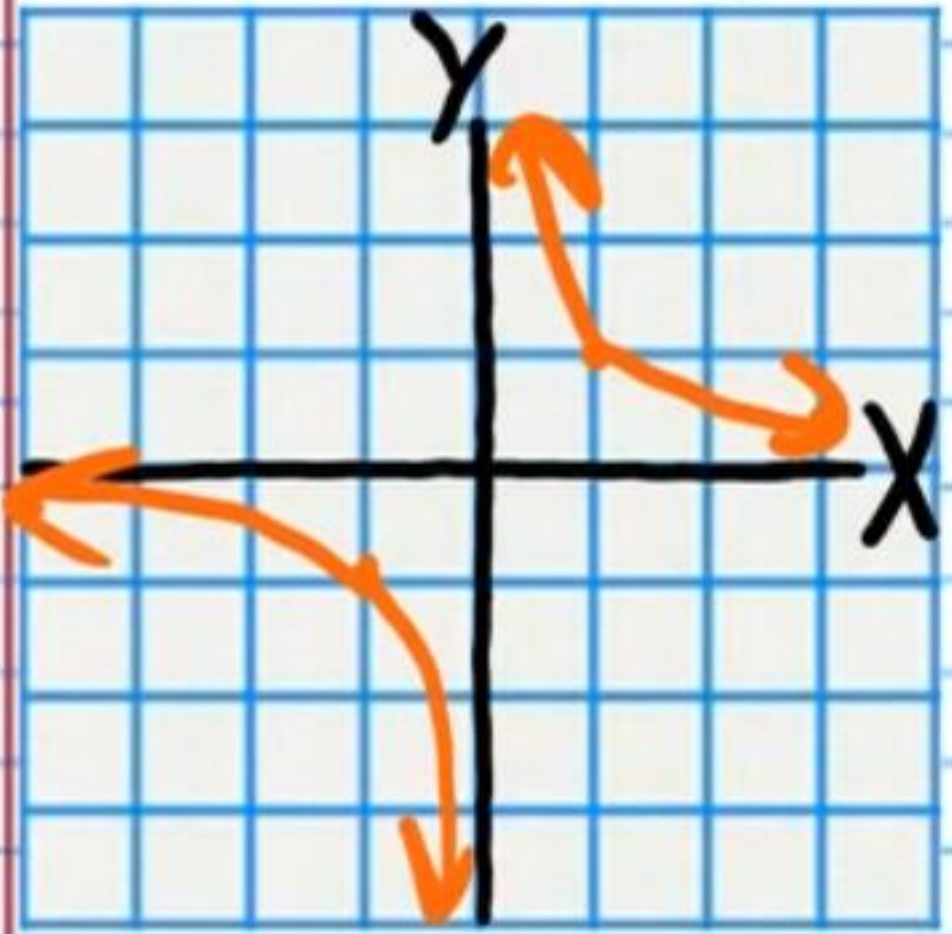
$x = \sqrt{4}, y = \frac{3}{2}$

$(2, \frac{3}{2})$

Eliminate the parameter

$$\begin{aligned} x &= \sqrt{t-2}, y = \frac{t}{4} \\ \downarrow & \qquad \qquad \downarrow \\ x &= \sqrt{4y-2} \quad t = 4y \end{aligned}$$

⑧ $x = t, y = \frac{1}{t}$



Point #1

$$\begin{aligned} t &= 1 \\ x &= 1, y = \frac{1}{1} \\ x &= 1, y = 1 \\ (1, 1) \end{aligned}$$

Point #2

$$\begin{aligned} t &= -1 \\ x &= -1, y = \frac{1}{-1} \end{aligned}$$

$x = -1, y = -1$

$(-1, -1)$

Point #4

$$\begin{aligned} t &= 3 \\ x &= 3, y = \frac{1}{3} \end{aligned}$$

$(3, \frac{1}{3})$

Point #3

$$\begin{aligned} t &= -4 \\ x &= -4, y = \frac{1}{-4} \end{aligned}$$

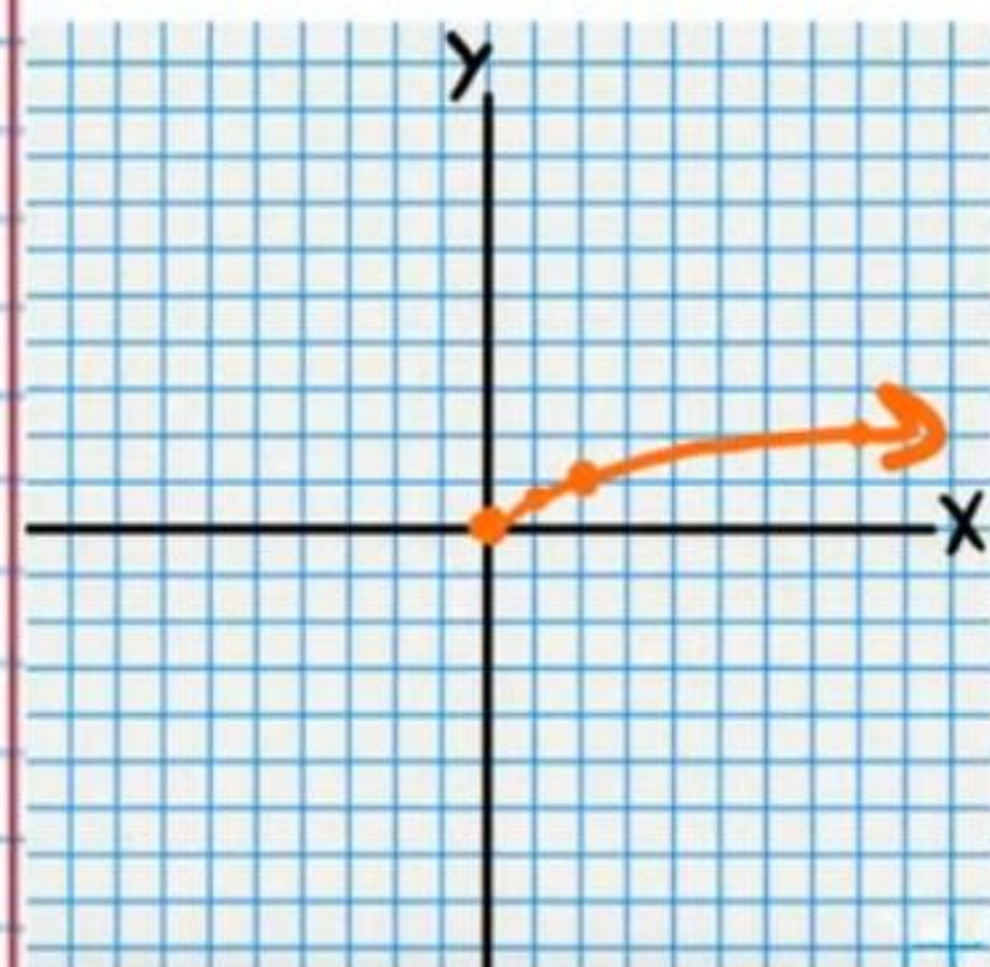
$(-4, -\frac{1}{4})$

Eliminate the parameter

$x = t, y = \frac{1}{t}$

$y = \frac{1}{x}$

• ⑨ $x = 2t, y = \sqrt{t}$



Point #1

$$\begin{aligned} t &= 0 \\ x &= 2(0), y = \sqrt{0} \\ x &= 0, y = 0 \\ \boxed{(0,0)} \end{aligned}$$

Point #3

$$\begin{aligned} t &= 1 \\ x &= 2(1), y = \sqrt{1} \\ x &= 2, y = 1 \\ \boxed{(2,1)} \end{aligned}$$

Point #4

$$\begin{aligned} t &= \frac{1}{2} \\ x &= 2\left(\frac{1}{2}\right), y = \sqrt{\frac{1}{2}} \\ x &= 1, y = \frac{\sqrt{2}}{2} \\ \boxed{\left(1, \frac{\sqrt{2}}{2}\right)} \end{aligned}$$

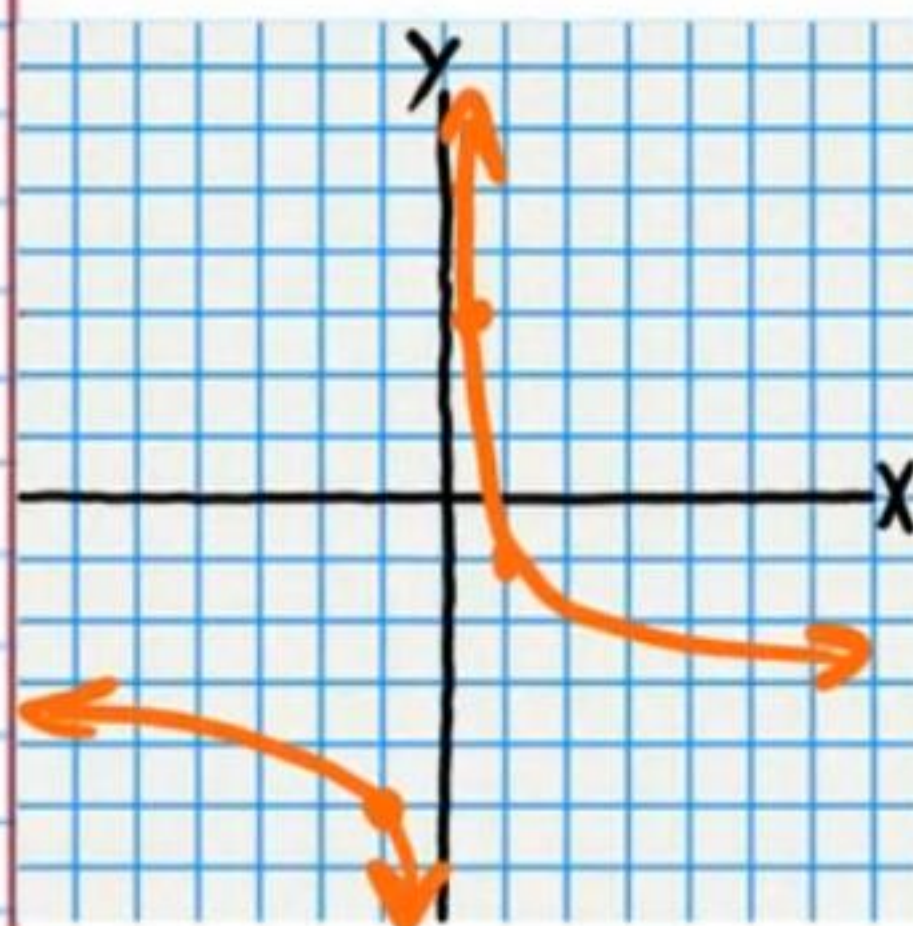
Point #2

$$\begin{aligned} t &= 4 \\ x &= 2(4), y = \sqrt{4} \\ x &= 8, y = 2 \\ \boxed{(8,2)} \end{aligned}$$

Eliminate the parameter

$$\begin{aligned} x &= 2t, y = \sqrt{t} \\ t &= \frac{x}{2} \\ y &= \sqrt{\frac{x}{2}} \\ y^2 &= \frac{x}{2} \end{aligned}$$

⑩ $x = \frac{2}{t+2}, y = t-1$



Point #1

$$\begin{aligned} t &= 0 \\ x &= \frac{2}{0+2}, y = 0-1 \\ x &= \frac{2}{2}, y = -1 \\ \boxed{(1,-1)} \end{aligned}$$

Point #2

$$\begin{aligned} t &= -4 \\ x &= \frac{2}{-4+2}, y = -4-1 \\ x &= -1, y = -5 \\ \boxed{(-1,-5)} \end{aligned}$$

Point #4

$$\begin{aligned} t &= 6 \\ x &= \frac{2}{6+2}, y = 6-1 \\ \boxed{\left(\frac{1}{4}, 5\right)} \end{aligned}$$

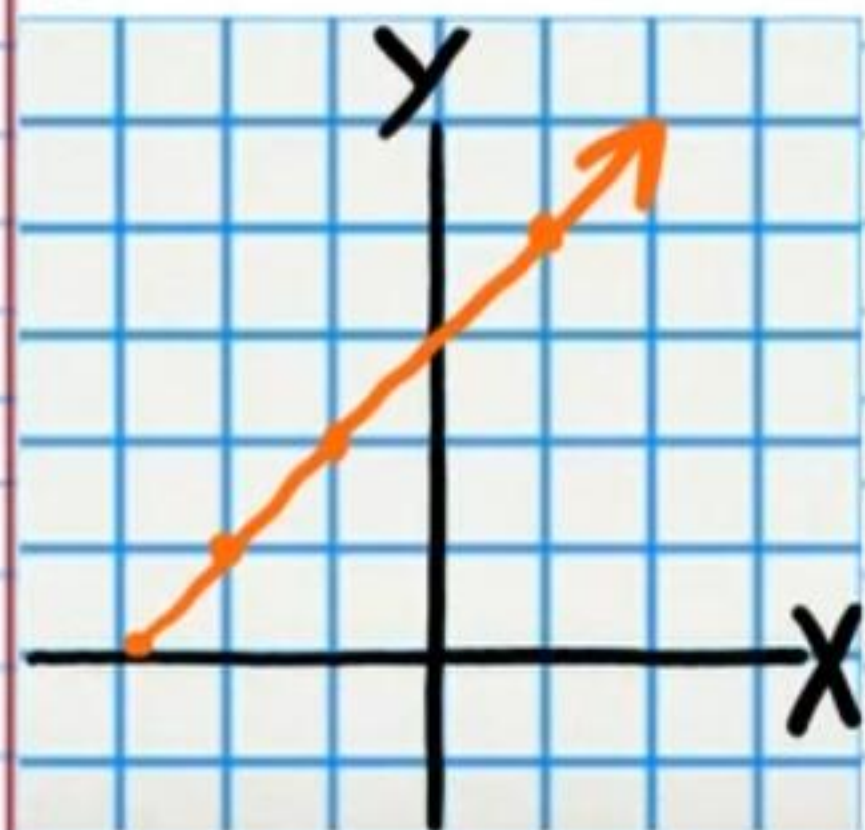
Point #3

$$\begin{aligned} t &= 4 \\ x &= \frac{2}{4+2}, y = 4-1 \\ x &= \frac{2}{6}, y = 3 \\ \boxed{\left(\frac{1}{3}, 3\right)} \end{aligned}$$

Eliminate the parameter

$$\begin{aligned} x &= \frac{2}{t+2}, y = t-1 \\ t &= y+1 \\ x &= \frac{2}{y+1+2} \\ x &= \frac{2}{y+3} \end{aligned}$$

• ⑪ $x = 2^t - 3, y = 2^t$ Point #1 Point #2



$t = 0$

$x = 2^0 - 3, y = 2^0$

$x = 1 - 3, y = 1$

$x = -2, y = 1$

$(-2, 1)$

$t = 1$

$x = 2^1 - 3, y = 2^1$

$x = 2 - 3, y = 2$

$x = -1, y = 2$

$(-1, 2)$

Point #3

$t = 2$

$x = 2^2 - 3, y = 2^2$

$x = 4 - 3, y = 4$

$x = 1, y = 4$

$(1, 4)$

Point #4

$t = -100$

$x = 2^{-100} - 3, y = 2^{-100}$

$x = \frac{1}{2^{100}} - 3, y = \frac{1}{2^{100}}$

$x \approx -3, y \approx 0$

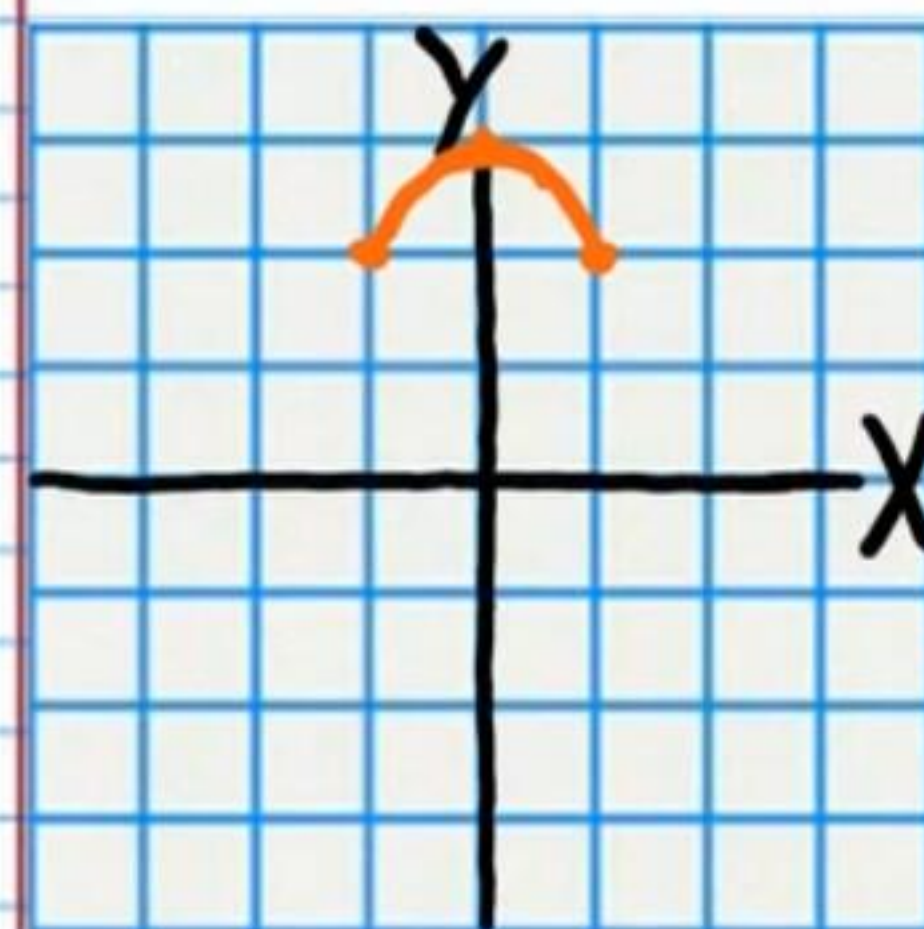
$(\approx -3, \approx 0)$

Eliminate the parameter

$x = 2^t - 3, y = 2^t$

$x = y - 3$

⑫ $x = \sin t, y = 3 - \sin^2 t$ Point #1



$t = 0^\circ$

$x = \sin 0^\circ, y = 3 - \sin^2 0^\circ$

$x = 0, y = 3 - 0^2$

$x = 0, y = 3$

$(0, 3)$

Point #2

$t = 90^\circ$

$x = \sin 90^\circ, y = 3 - \sin^2 90^\circ$

$x = 1, y = 3 - 1^2$

$x = 1, y = 2$

$(1, 2)$

Point #4

$t = 30^\circ$

$x = \sin 30^\circ, y = 3 - \sin^2 30^\circ$

$x = \frac{1}{2}, y = 3 - \left(\frac{1}{2}\right)^2$

$\left(\frac{1}{2}, 2\frac{3}{4}\right)$

Point #3

$t = -90^\circ$

$x = \sin(-90^\circ), y = 3 - \sin^2(-90^\circ)$

$x = -1, y = 3 - (-1)^2$

$x = -1, y = 3 - 1$

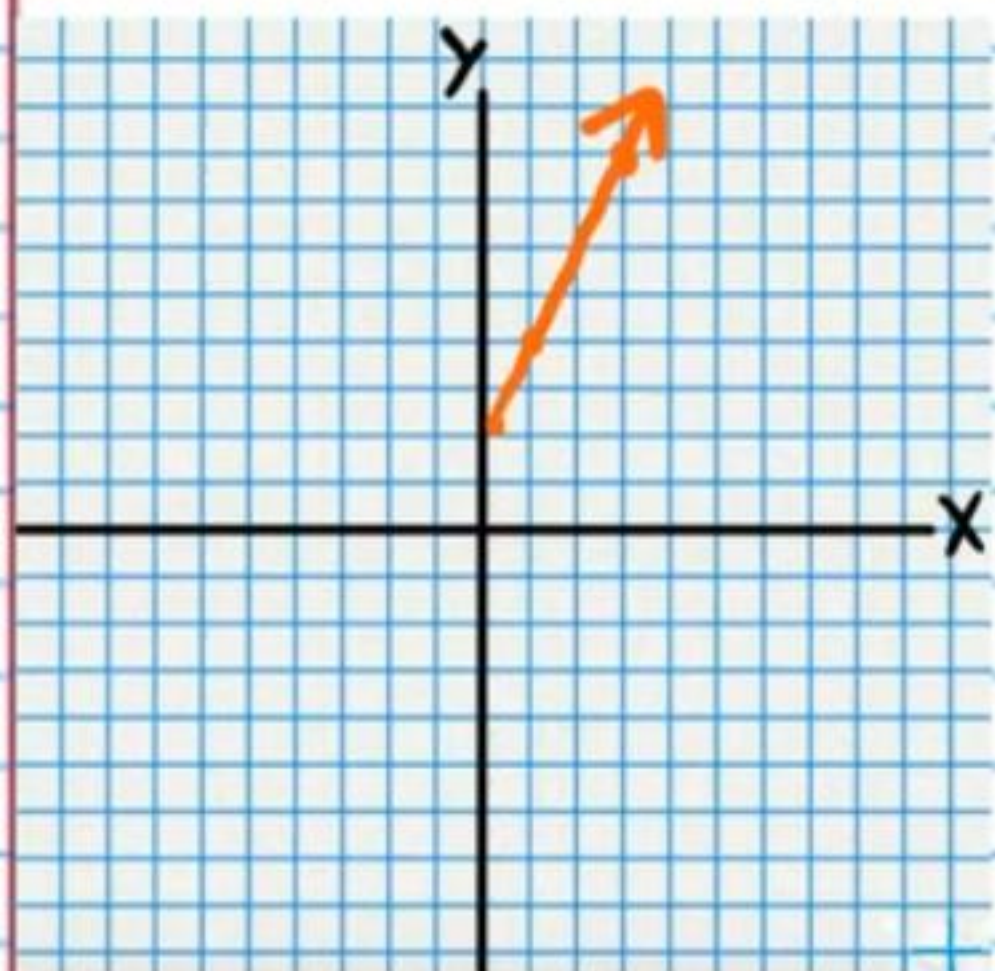
$(-1, 2)$

Eliminate the parameter

$x = \sin t, y = 3 - \sin^2 t$

$y = 3 - x^2$

• ⑬ $X = 3^t, Y = 2(3^t) + 2$



Point #1

$t = 0$

$X = 3^0, Y = 2(3^0) + 2$

$X = 1, Y = 2(1) + 2$

$X = 1, Y = 4$

$(1, 4)$

Point #2

$t = 1$

$X = 3^1, Y = 2(3^1) + 2$

$X = 3, Y = 2(3) + 2$

$(3, 8)$

Point #4

$t = \frac{1}{2}$

$X = 3^{\frac{1}{2}}, Y = 2(3^{\frac{1}{2}}) + 2$

$(\sqrt{3}, 2\sqrt{3} + 2)$

Point #1

$t = 0$

$X = 3^0, Y = 2(3^0) + 2$

$X = 1, Y = 2(1) + 2$

$X = 1, Y = 4$

$(1, 4)$

Point #3

$t = -3$

$X = 3^{-3}, Y = 2(3^{-3}) + 2$

$X = \frac{1}{3^3}, Y = 2(\frac{1}{3^3}) + 2$

$X = \frac{1}{27}, Y = 2(\frac{1}{27}) + 2$

$(\frac{1}{27}, 2\frac{2}{27})$

Eliminate the parameter

$X = 3^t, Y = 2(3^t) + 2$

$Y = 2X + 2$

⑭ $X = \cos^3 t, Y = \cos t + 1$

Point #1

$t = 0^\circ$

$X = \cos^3 0^\circ, Y = \cos 0^\circ + 1$

$X = 1, Y = 1 + 1$

$X = 1, Y = 2$

$(1, 2)$

Point #2

$t = 90^\circ$

$X = \cos^3 90^\circ, Y = \cos 90^\circ + 1$

$X = (\cos 90^\circ)^3, Y = 0 + 1$

$X = 0^3, Y = 1$

$(0, 1)$

Point #4

$t = 60^\circ$

$X = \cos^3 60^\circ, Y = \cos 60^\circ + 1$

$X = (\frac{1}{2})^3, Y = \frac{1}{2} + 1$

$(\frac{1}{8}, 1\frac{1}{2})$

Point #3

$t = 180^\circ$

$X = \cos^3(180^\circ), Y = \cos 180^\circ + 1$

$X = (-1)^3, Y = -1 + 1$

$X = -1, Y = 0$

$(-1, 0)$

Eliminate the parameter

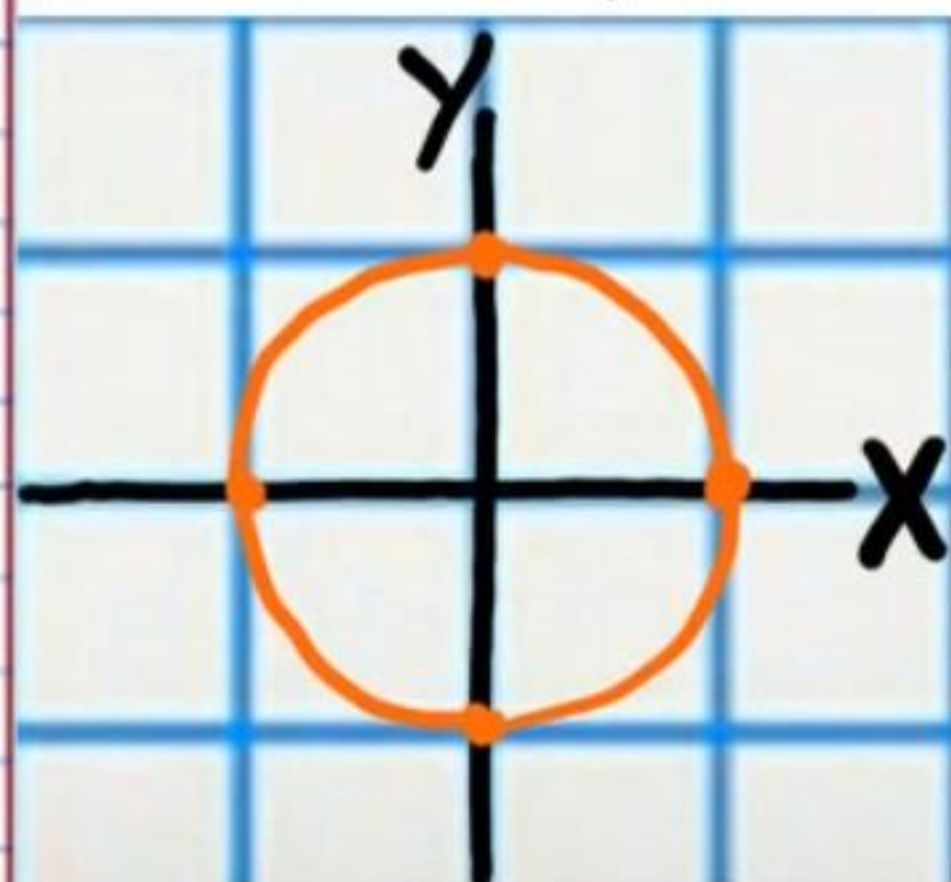
$X = \cos^3 t, Y = \cos t + 1$

$X = (Y - 1)^3$

$\cos t = Y - 1$

$$\sin^2 t + \cos^2 t = 1$$

• ⑮ $x = \cos t, y = \sin t$



Point #1

$$t = 0^\circ$$

$$x = \cos 0^\circ, y = \sin 0^\circ$$

$$x = 1, y = 0$$

$$\boxed{(1, 0)}$$

Point #2

$$t = 90^\circ$$

$$x = \cos 90^\circ, y = \sin 90^\circ$$

$$x = 0, y = 1$$

$$\boxed{(0, 1)}$$

Point #3

$$t = -90^\circ$$

$$x = \cos(-90^\circ), y = \sin(-90^\circ)$$

$$x = 0, y = -1$$

$$\boxed{(0, -1)}$$

Point #4

$$t = 180^\circ$$

$$x = \cos 180^\circ, y = \sin 180^\circ$$

$$x = -1, y = 0$$

$$\boxed{(-1, 0)}$$

Eliminate the parameter

$$x = \cos t, y = \sin t$$

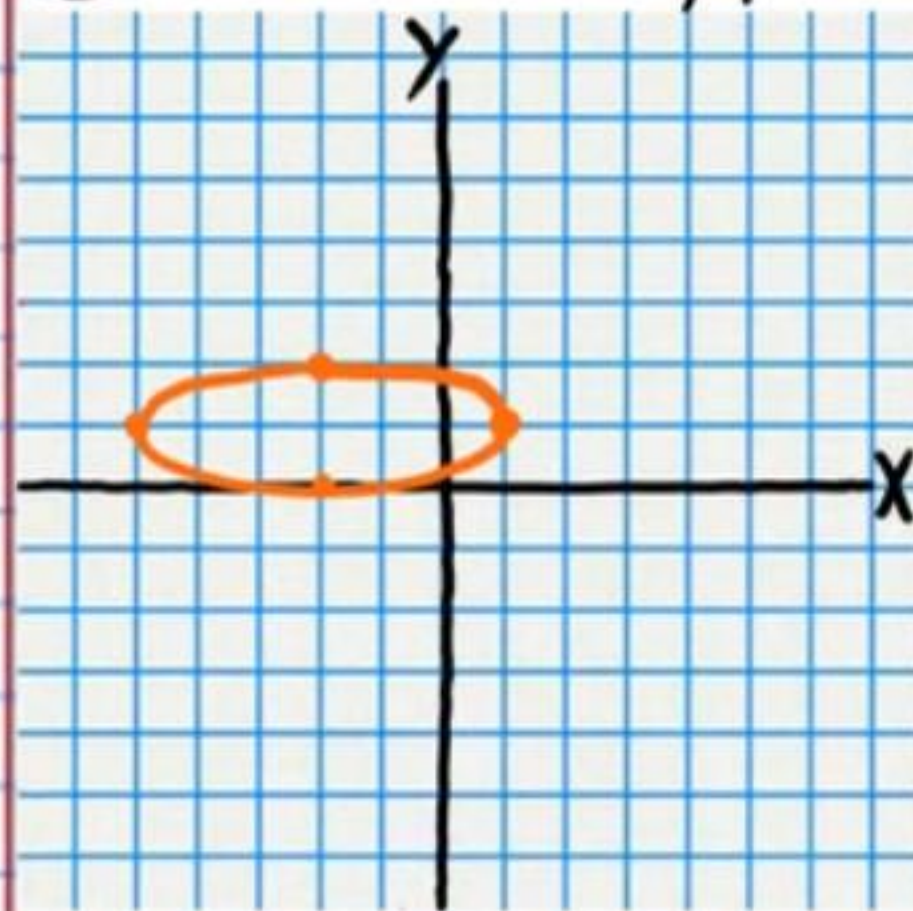
$$\sin^2 t + \cos^2 t = 1$$

$$(\sin t)^2 + (\cos t)^2 = 1$$

$$(y)^2 + (x)^2 = 1$$

$$\boxed{y^2 + x^2 = 1}$$

⑯ $x = 3\sin t - 2, y = \cos t + 1$ Point #1



$$t = 0^\circ$$

$$x = 3\sin 0^\circ - 2, y = \cos 0^\circ + 1$$

$$x = 3(0) - 2, y = 1 + 1$$

$$\boxed{(-2, 2)}$$

Point #2

$$t = 90^\circ$$

$$x = 3\sin 90^\circ - 2, y = \cos 90^\circ + 1$$

$$x = 3(1) - 2, y = 0 + 1$$

$$\boxed{(1, 1)}$$

Point #4

$$t = 180^\circ$$

$$x = 3\sin 180^\circ - 2, y = \cos 180^\circ + 1$$

$$x = 3(0) - 2, y = -1 + 1$$

$$\boxed{(-2, 0)}$$

Point #3

$$t = -90^\circ$$

$$x = 3\sin(-90^\circ) - 2, y = \cos(-90^\circ) + 1$$

$$x = 3(-1) - 2, y = 0 + 1$$

$$\boxed{(-5, 1)}$$

Eliminate the parameter

$$x = 3\sin t - 2 \quad \left\{ \begin{array}{l} y = \cos t + 1 \end{array} \right.$$

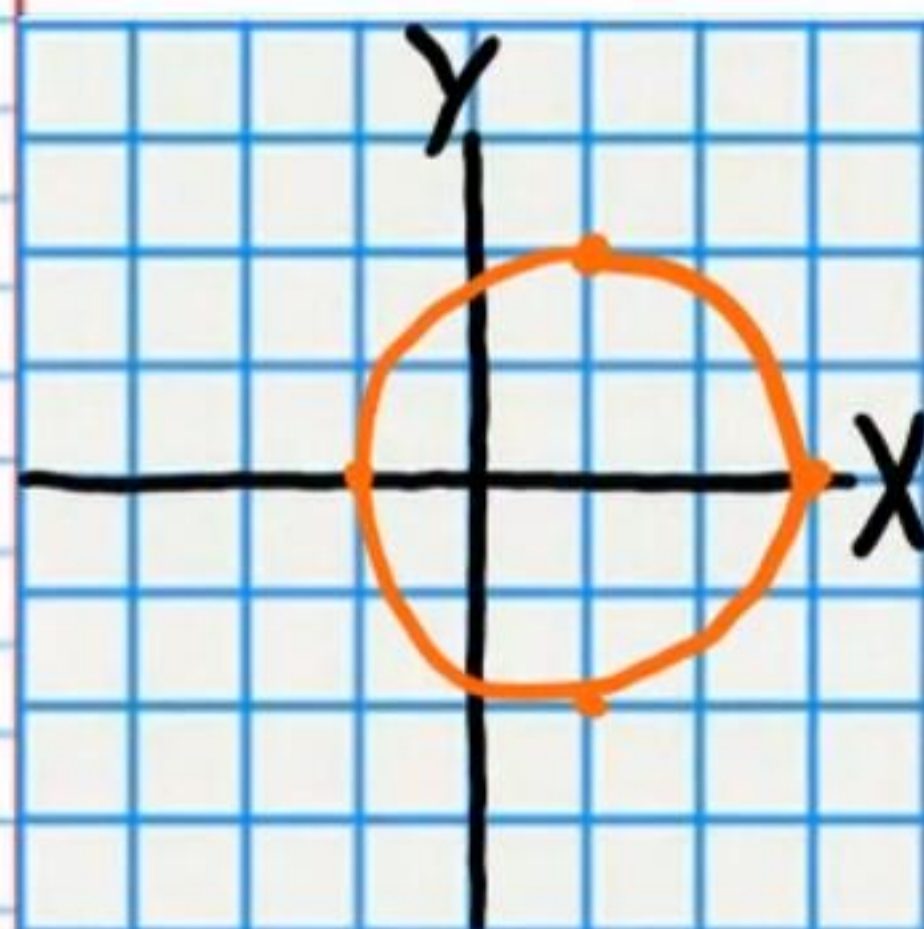
$$\frac{x+2}{3} = \sin t$$

$$\sin^2 t + \cos^2 t = 1$$

$$\left(\frac{x+2}{3}\right)^2 + (y-1)^2 = 1$$

$$\boxed{\frac{(x+2)^2}{9} + (y-1)^2 = 1}$$

• (17) $x = 2\sin t + 1, y = 2\cos t$ Point #1



$t = 0^\circ$

$x = 2\sin 0^\circ + 1, y = 2\cos 0^\circ$

$x = 1, y = 2$

$(1, 2)$

Point #2

$t = 90^\circ$

$x = 2\sin 90^\circ + 1, y = 2\cos 90^\circ$

$(3, 0)$

Point #3

$t = -90^\circ$

$x = 2\sin(-90^\circ) + 1, y = 2\cos(-90^\circ)$

$(-1, 0)$

Point #4

$t = 180^\circ$

$x = 2\sin 180^\circ + 1, y = 2\cos 180^\circ$

$(1, -2)$

Eliminate the parameter

$x = 2\sin t + 1, y = 2\cos t$

$x - 1 = 2\sin t \quad \left\{ \begin{array}{l} \frac{y}{2} = \cos t \end{array} \right.$

$\frac{x-1}{2} = \sin t$

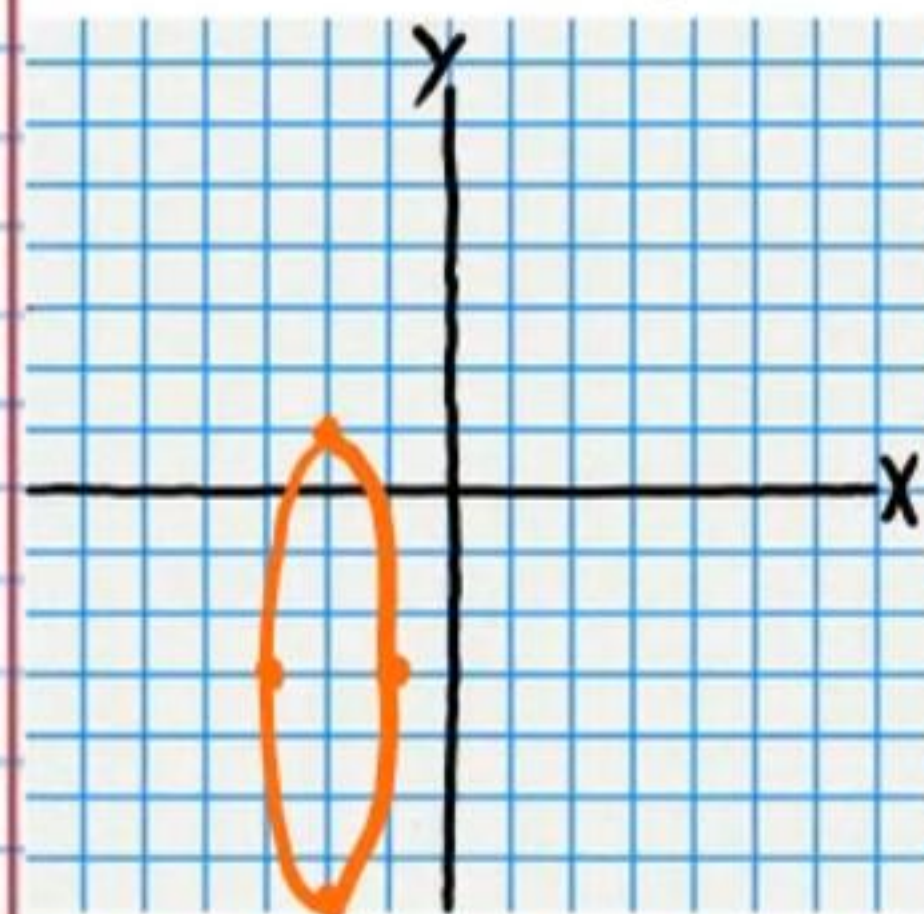
$\sin^2 t + \cos^2 t = 1$

$\left(\frac{x-1}{2}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$

$\frac{(x-1)^2}{4} + \frac{y^2}{4} = 1$

$(x-1)^2 + y^2 = 4$

⑮ $x = \sin t - 2, y = 4 \cos t - 3$ **Point #1**



$t = 0^\circ$
 $x = \sin 0^\circ - 2, y = 4 \cos 0^\circ - 3$

$(-2, 1)$

Point #2

$t = 90^\circ$
 $x = \sin 90^\circ - 2, y = 4 \cos 90^\circ - 3$

$(-1, -3)$

Point #3

$t = -90^\circ$
 $x = \sin(-90^\circ) - 2, y = 4 \cos(-90^\circ) - 3$

$(-3, -3)$

Eliminate the parameter

$x = \sin t - 2, y = 4 \cos t - 3$

$\sin t = x + 2 \quad \cos t = \frac{y + 3}{4}$

$\sin^2 t + \cos^2 t = 1$

$(x + 2)^2 + \left(\frac{y + 3}{4}\right)^2 = 1$

$(x + 2)^2 + \frac{(y + 3)^2}{16} = 1$

Point #4

$t = 180^\circ$
 $x = \sin 180^\circ - 2, y = 4 \cos 180^\circ - 3$

$(-2, -7)$

Creating Parametric Equations

Create parametric equations for x and y using t as the parameter.

• ⑲ $y = x^2 + 4x - 1$

$x = t, y = t^2 + 4t - 1$

⑳ $y^3 = x$

$x = t^3, y = t$

㉑ $y + 2 = 4^{x+1}$

$x = t, y + 2 = 4^{t+1}$

$x = t, y = 4^{t+1} - 2$

• ㉒ $y = \frac{1}{5x}$

$x = t, y = \frac{1}{5t}$

㉓ $\frac{3x^2}{2} = y$

$x = t, y = \frac{3t^2}{2}$

㉔ $x = 2^y$

$x = 2^t, y = t$